

EDUCATION

Emory University, Atlanta, GA

Aug 2022 – May 2026

- Bachelor of Science, GPA: 3.72 / 4.00, Honors Thesis: Probing Artificial Memory (Advisor: Li Xiong)
- Majors in *Physics (Engineering Science Concentration)* and *Computer Science & Economics Joint Program (Highest Honors)*
- Relevant Coursework: Machine Learning, Probability and Statistics, Artificial Intelligence, Computational Modeling, Mathematical Methods for Scientists & Engineers, Data Mining, Data Structures & Algorithms, Quantum Mechanics

Canadian International School of Hong Kong, Hong Kong

May 2022

- *IB Diploma*, IB Score: 41/45, Honor Roll, Maple Leaf Award and *OSSD Diploma*, OSSD Score: 96%, OSSD Scholars Award

PROFESSIONAL EXPERIENCE

Emory University AIMS Lab, *Research Assistant (Advisor: Li Xiong)*, Atlanta, GA

May 2025 – Present

- Co-first authored a paper studying machine learning privacy, developing a membership inference attack (MIA) on large language and visual language models (VLLM) improving upon baselines by 18% (**Under Review at ICLR 2027**)
- Authored thesis developing a geometric account of VLLM memorization under low-noise perturbation, extending the attack to medical models and various modalities including MRI, endoscopy, and microscopy data (**Published Thesis**)
- Initiated a follow-up project applying information geometry to perturbation-response landscapes to give a mechanistic account of the MIA signals and whether memorization leaves geometric signatures in VLLM representations (**ongoing**)
- Exploring empirical scaling laws for MIAs to predict the success of well-resourced attackers from low-cost defender audits, making privacy risk measurable for models too large to audit directly (**ongoing**)

Emory University Physics Department, *Undergraduate Researcher*, Atlanta, GA

Aug 2023 – May 2025

- Built and refined a magneto-optic Kerr effect (MOKE) rig with lock-in to measure hysteresis loops in low-anisotropy magnetic multilayers fabricated by magnetron sputtering, iterating on samples and setup to improve signal-to-noise
- Restored and calibrated the lab's atomic force microscope and served as its primary trainer, teaching graduate students instrument operation and sample preparation and surface characterization of thin-film samples
- Automated instrument control in LabVIEW and wrote Python pipelines for signal processing, curve fitting and statistical analysis of hysteresis data, improving measurement reproducibility across sample batches

Tripple HHH. Inc., *Foreign Investment Strategy Intern*, Toronto, Canada

Nov 2023 – Jan 2024

- Built IFRS capital budgeting models (payback, MIRR) to evaluate 5 London properties, leading to the purchase of 1
- Projected leveraged returns for London properties against the company's WACC and wrote investment proposals recommending acquisitions to senior management based on risk-adjusted comparisons of projected returns
- Corresponded with overseas agents to review due diligence and coordinate the transaction through to closing

Richland Worldwide Ltd., *Junior Research Analyst Intern*, Hong Kong

Jun 2023 – Aug 2023

- Valued NASDAQ-listed technology and financial-services firms using CAPM risk assessment, dividend-discount and bond valuation models, presenting investment pitches on 20 listed companies to senior management
- Wrote Python scripts automating the processing of 10 datasets and built data visualizations for pitch decks

LEADERSHIP & COMMUNITY INVOLVEMENT

Emory Robotics, *Treasurer, then President*, Atlanta, GA

Aug 2022 – Aug 2025

- Led the team to its first VEX robotics competition, winning 3 qualification matches and reaching the final 16 teams
- Managed development of 2 robots (\$10,000 total), overseeing C++ pure-pursuit path following control on VEX PROS
- Raised 3.5x previous year's annual budget within 6 months as club treasurer preparing and defending funding proposals

BASE Design Organization, Founder, Hong Kong

Jun 2020 – Jun 2022

- Founded a student venture designing and manufacturing consumer goods, leading 30 members across 3 teams
- Provided consulting, products and learning workshops to local elderly homes and NGOs as community service

ADDITIONAL INFORMATION

Research Interests: Memorization of vision LLMs, Privacy and trustworthy ML, Physics inspired theory of deep learning

Programming: Python (PyTorch, NumPy, Pandas, Scikit-learn), Java, LaTeX, Git, LabVIEW, Linux / GPU cluster computing

Languages: English (Fluent), Mandarin (Proficient), Cantonese (Proficient), French (Basic)

Other Interests: Aircraft Modeling, CAD design and 3D printing, Football, Badminton, Boxing, Cooking